300mW, 2.4V - 39V Surface Mount Zener Diode

FEATURES

- AEC-Q101 qualified
- Zener voltage range selection: 2.4V to 39V
- V_Z tolerance selection of ± 5%
- Ideally suited for automated assembly processes
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

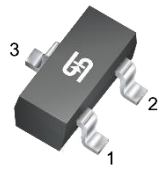
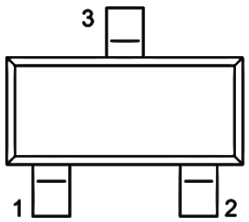
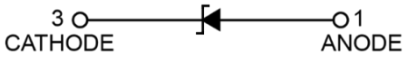
KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
P_D	300	mW
V_Z	2.4 - 39	V
$T_{J\ MAX}$	150	°C
V_F at I_F=10mA	0.9	V
Configuration	Single die	

APPLICATIONS

- General regulation functions

MECHANICAL DATA

- Case: SOT-23
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Weight: 8.00mg (approximately)

PACKAGE: SOT-23	PIN CONFIGURATION	CIRCUIT DIAGRAM
		

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

PARAMETER	SYMBOL	VALUE	UNIT
Power dissipation(1)	P_D	300	mW
Forward voltage @ I_F=10mA	V_F	0.9	V
Junction temperature	T_J	-55 to +150	°C
Storage temperature	T_{STG}	-55 to +150	°C

Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance(1)	$R_{\Theta JA}$	417	°C/W

Thermal Performance Note:

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PART NUMBER	DEVICE MARKING	ZENER VOLTAGE				REGULAR IMPEDANCE			LEAKAGE CURRENT		TEMPERATURE COEFFICIENT	
		V_Z @ I_{ZT}			I_{ZT}	Z_{ZT} @ I_{ZT}	Z_{ZK} @ I_{ZK}	I_{ZK}	I_R @ V_R		@ I_{ZT}	
		V			mA	Ω	Ω	mA	μA	V	mV/°C	
		Min	Nom	Max		Max	Max		Max		Min	Max
BZX84C2V4H	Z11	2.2	2.4	2.6	5	100	600	1.0	50	1.0	-3.5	0
BZX84C2V7H	Z12	2.5	2.7	2.9	5	100	600	1.0	20	1.0	-3.5	0
BZX84C3V0H	Z13	2.8	3.0	3.2	5	95	600	1.0	10	1.0	-3.5	0
BZX84C3V3H	Z14	3.1	3.3	3.5	5	95	600	1.0	5	1.0	-3.5	0
BZX84C3V6H	Z15	3.4	3.6	3.8	5	90	600	1.0	5	1.0	-3.5	0
BZX84C3V9H	Z16	3.7	3.9	4.1	5	90	600	1.0	3	1.0	-3.5	0
BZX84C4V3H	Z17	4.0	4.3	4.6	5	90	600	1.0	3	1.0	-3.5	0.0
BZX84C4V7H	Z1	4.4	4.7	5.0	5	80	500	1.0	3	2.0	-3.5	0.2
BZX84C5V1H	Z2	4.8	5.1	5.4	5	60	480	1.0	2	2.0	-2.7	1.2
BZX84C5V6H	Z3	5.2	5.6	6.0	5	40	400	1.0	1	2.0	-2.0	2.5
BZX84C6V2H	Z4	5.8	6.2	6.6	5	10	150	1.0	3	4.0	0.4	3.7
BZX84C6V8H	Z5	6.4	6.8	7.2	5	15	80	1.0	2	4.0	1.2	4.5
BZX84C7V5H	Z6	7.0	7.5	7.9	5	15	80	1.0	1	5.0	2.5	5.3
BZX84C8V2H	Z7	7.7	8.2	8.7	5	15	80	1.0	0.7	5.0	3.2	6.1
BZX84C9V1H	Z8	8.5	9.1	9.6	5	15	100	1.0	0.5	6.0	3.8	7.0
BZX84C10H	Z9	9.4	10	10.6	5	20	150	1.0	0.2	7.0	4.5	8.0
BZX84C11H	Y1.	10.4	11	11.6	5	20	150	1.0	0.1	8.0	5.4	9.0
BZX84C12H	Y2.	11.4	12	12.7	5	25	150	1.0	0.1	8.0	6.0	10.0
BZX84C13H	Y3	12.4	13	14.1	5	30	170	1.0	0.1	8.0	7.0	11.0
BZX84C15H	Y4	13.8	15	15.6	5	30	200	1.0	0.1	10.5	9.2	13.0
BZX84C16H	Y5	15.3	16	17.1	5	40	200	1.0	0.1	11.2	10.4	14.0
BZX84C18H	Y6	16.8	18	19.1	5	45	225	1.0	0.1	12.6	12.4	16.0
BZX84C20H	Y7	18.8	20	21.2	5	55	225	1.0	0.1	14.0	14.4	18.0
BZX84C22H	Y8	20.8	22	23.3	5	55	250	1.0	0.1	15.4	16.4	20.0
BZX84C24H	Y9	22.8	24	25.6	5	70	250	1.0	0.1	16.8	18.4	22.0
BZX84C27H	Y10	25.1	27	28.9	2	80	300	0.5	0.1	18.9	21.4	25.3
BZX84C30H	Y11	28.0	30	32.0	2	80	300	0.5	0.1	21.0	24.4	29.4
BZX84C33H	Y12	31.0	33	35.0	2	80	325	0.5	0.1	23.1	27.4	33.4
BZX84C36H	Y13	34.0	36	38.0	2	90	350	0.5	0.1	25.2	30.4	37.4
BZX84C39H	Y14	37.0	39	41.0	2	130	350	0.5	0.1	27.3	33.4	41.2

ORDERING INFORMATION

ORDERING CODE(1)	PACKAGE	PACKING
BZX84CxxxH RFG	SOT-23	3,000 / 7" Tape & Reel

Notes:

1. "xxx" defines voltage from 2.4V(BZX84C2V4H) to 39V(BZX84C39H)

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Power Dissipation Curve

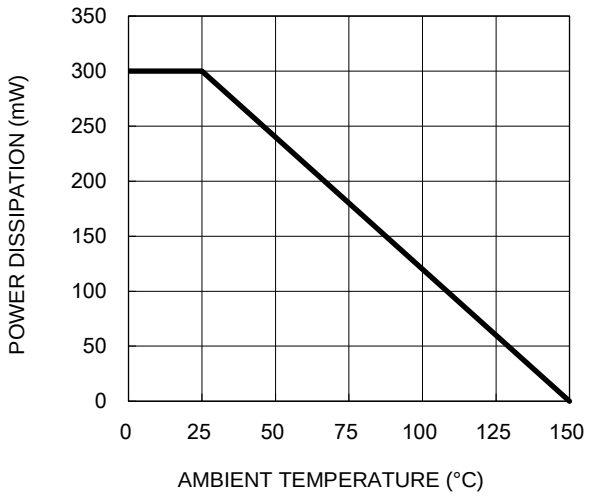


Fig.2 Zener Breakdown Characteristics

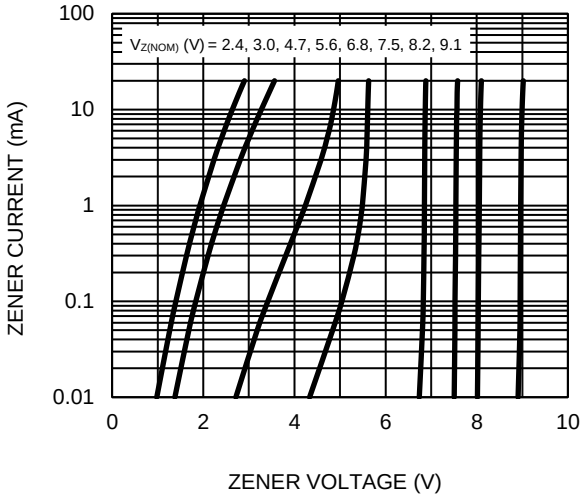


Fig.3 Zener Breakdown Characteristics

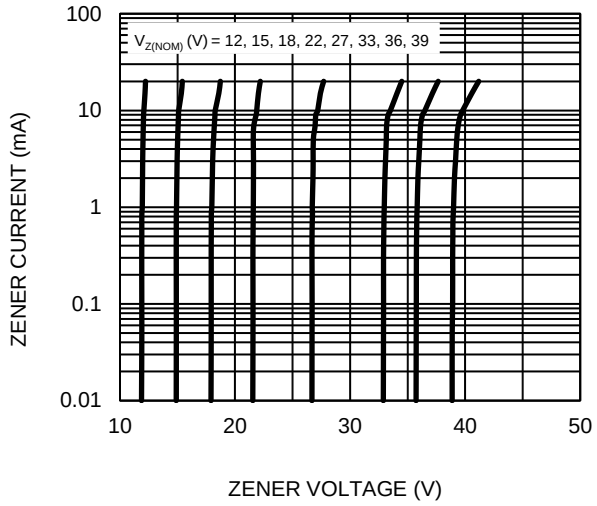


Fig.4 Typical Junction Capacitance

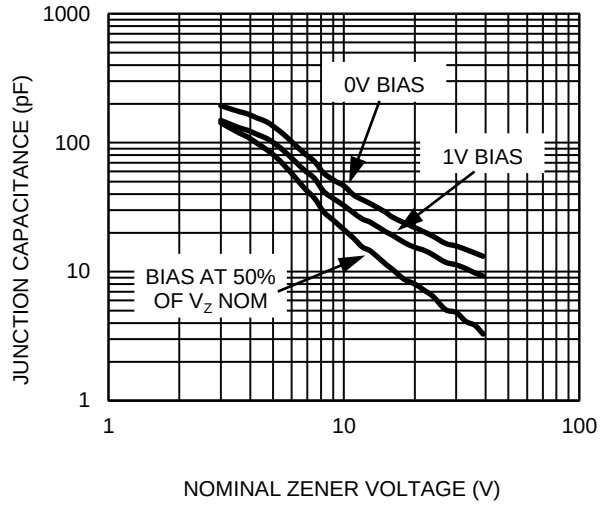
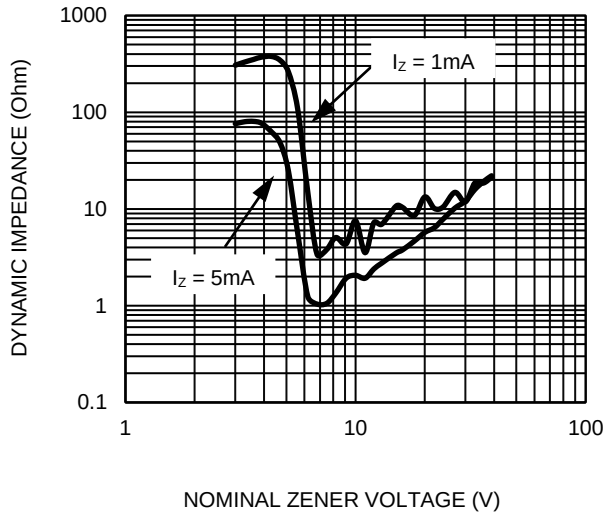
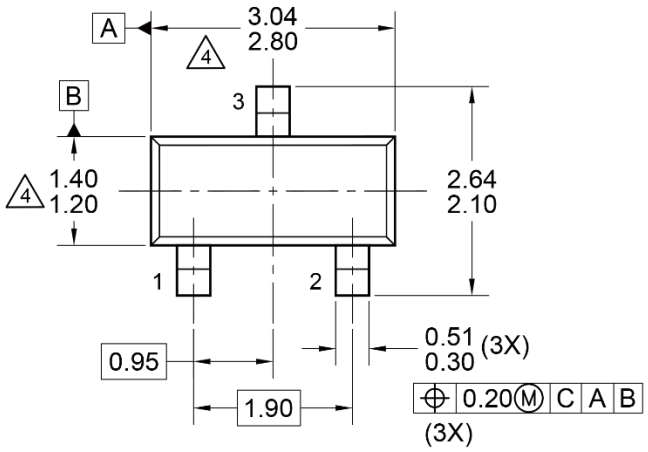


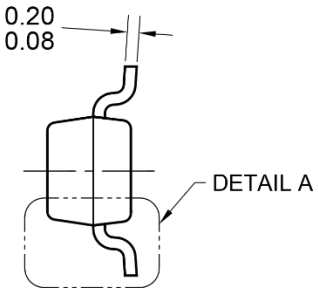
Fig.5 Effect of Zener Voltage on Impedance

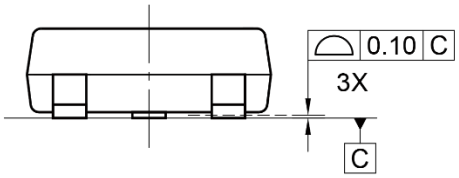


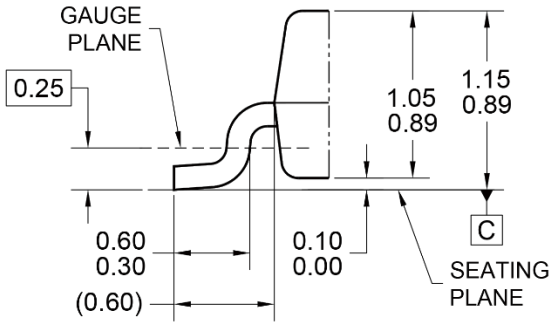
PACKAGE OUTLINE DIMENSIONS

SOT-23

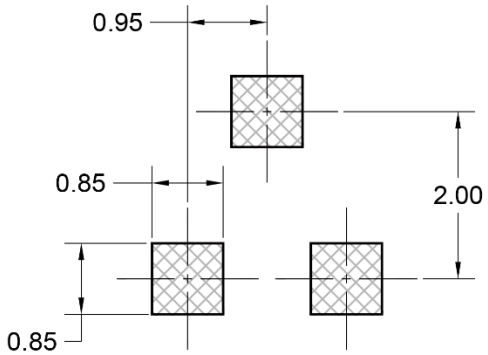




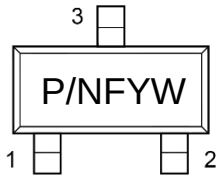




DETAIL A, ROTATED -90°
(SCALE 2:1)



SUGGESTED PAD LAYOUT



MARKING DIAGRAM

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: JEDEC TO-236, ISSUE H, VARIATION AA.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-SOT23JEDEC-104 REV B.

P/N = Device marking
F = Factory code
Y = Year code
W = Bi-Week code (A~Z)

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